Conditional Probability

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Independence of an event :- Let A & B are the two event defined on Ω said to be independent if it satisfy the following condition.

$$P(A \cap B) = P(A) \cdot P(B)$$

Ex:- $\Omega = \{1, 2, 3, 4, 5, 6\}$

A : An even no. occur

$A = \{2, 4, 6\}$

B : No. greater than 4 occurs

$B = \{5, 6\}$

$A \cap B = \{6\}$

$P(A \cap B) = \frac{1}{6}$ —— (1)

$P(A) = \frac{3}{6} = \frac{1}{2}$

$P(B) = \frac{2}{6} = \frac{1}{3}$

$P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$ —— (2)

from (1) and (2)

$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{6}$

$\perp\!\!\!\perp$ } independent

Theorem 1: Suppose A and B are two events defined on Ω. If A & B are independent then
(i) A & B' are independent
(ii) A' & B are independent
(iii) A' & B' are independent

(i) If A & B' are independent then,
$$P(A \cap B') = P(A) \cdot P(B')$$

$\therefore$ L.H.S
$$= P(A \cap B')$$
$$= P(A) - P(A \cap B)$$
$$= P(A) - P(A) \cdot P(B)$$
$$= P(A)\,[1 - P(B)]$$
$$= P(A) \cdot P(B')$$

Hence proved

(ii) If A' & B are independent then.
$$P(A' \cap B) = P(A') \cdot P(B)$$
L.H.S
$$= P(A' \cap B)$$
$$= P(B) - P(A \cap B)$$
$$= P(B) - [P(A) \cdot P(B)]$$
$$= P(B) \cdot [1 - P(A)]$$
$$= P(B)\,[P(A')]$$
$$= P(B) \cdot P(A')$$

is defined then

(iii) A' & B' are independent then;

$P(A' \cap B') = P(A') \cdot P(B')$

$L.H.S = P(A' \cap B')$

$= P(A \cup B)'$

$= [1 - P(A \cup B)]$

$= 1 - [P(A) + P(B) - P(A \cap B)]$

$= 1 - P(A) - P(B) + P(A \cap B)$

$\cancel{= [1 - P(A)] - P(B)\ [1 - P(}\ [$

$= [1 - P(A)] - P(B) + P(A) \cdot P(B)$

$= [1 - P(A)] - P(B) \cdot [1 - P(A)]$

$= [1 - P(A)]\,[1 - P(B)]$

$= P(A') \cdot P(B')$

$= R.H.S$

ed

Ex 1] A & B are independent with $P(A) = 0.5$ $P(B) = 0.4$ find (i) $P(A \cup B)$ (ii) $P(A \cap B')$ (iii) $P(A' \cap B)$ (iv) $P(A' \cap B')$

$\Rightarrow$ Since A & B are independent,

$\therefore P(A \cap B) = P(A) \cdot P(B)$

$P(A \cap B) = (0.5)(0.4)$

$\boxed{P(A \cap B) = 2.0}$

$$\begin{array}{r} 0.5 \\ 0.4 \\ \hline 2.0 \end{array}$$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.5 + 0.4 - 2.0$

$= 0.9 - 2.0$

(ii) $P(A \cap B') = P(A) - P(A \cap B)$

$= 0.5 -$

A & B are ||

$P(A \cap B) = P(A) \cdot P(B)$ —— (1)

(iv) $P(A' \cap B') = P(A \cup B)'$

$= [1 - P(A \cup B)]$

$= 1 - [P(A) + P(B) - P(A \cap B)]$

$= [1 - P(A) - P(B) + P(A \cap B)]$

~~$= [1 - P(A)] - $~~

$= 1 - [P(A) - P(B) + P(A) \cdot P(B)]$

$= 1 - [0.9 - 0.20]$

$= 1 - 0.7$

$P(A' \cap B') = 0.3$

Ex 1] Suppose a card is drawn at random from a well shuffled pack of 52 cards.
Let event A = getting a spade card.
B = getting a king
A & B is independent prove it.

⇒ Given:-

A = getting spade cards
B = getting king cards

$$P(A) = \frac{n(A)}{n(S)} = \frac{^{13}C_1}{^{52}C_1} = \frac{13}{52}$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{^{4}C_1}{^{52}C_1} = \frac{4}{52}$$

$$P(A) \cdot P(B) = \frac{\cancel{13}^{1}}{\cancel{52}_{13}} \cdot \frac{\cancel{4}^{1}}{52}$$

$$\boxed{P(A) \cdot P(B) = \frac{1}{52}} \quad \text{—— (1)}$$

$$\boxed{P(A \cap B) = \frac{1}{52}} \quad \text{—— (2)}$$

∴ From (1) and (2) we can say.

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{52}$$

Thus, A and B are independent (II)

Ex 2]

⇒

(i)

(ii)

Ex 2] Suppose A & B are 2 events defined on Ω. If $P(A) = 0.8$, $P(A \cup B) = 0.9$ & $P(B) = x$, find the value of x if A & B are (i) independent (ii) mutually exclusive.

$\Rightarrow$ Given that:

$P(A \cup B) = 0.9$

$P(A) + P(B) - P(A \cap B) = 0.9$

$0.8 + x - P(A \cap B) = 0.9$ —— (I)

(i) Independent:-

$P(A \cap B) = P(A) \cdot P(B)$ —— (II)

from (I) and (II)

$0.8 + x - P(A) \cdot P(B) = 0.9$

$0.8 + x - (0.8)(x) = 0.9$

$0.8 + x - 0.8x = 0.9$

$0.2x = 0.1$

$$\boxed{x = \frac{0.1}{0.2} = \frac{1}{2} = 0.5}$$

$$\boxed{\therefore P(B) = x = 0.5}$$

(ii) Mutually exclusive:-

$\therefore$ If A & B are mutually exclusive

$\therefore P(A \cap B) = 0$

eq^n (I) becomes

$0.8 + x - 0 = 0.9$

$0.8 + x = 0.9$

$$\boxed{x = 0.1 = P(B)}$$

Mutual Independence :-

Let A, B & C are any three events defined on Ω said to be mutually independent or completely independent if satisfies the following conditions:-

(I) $P(A \cap B) = P(A) \cdot P(B)$

(II) $P(B \cap C) = P(B) \cdot P(C)$

(III) $P(A \cap C) = P(A) \cdot P(C)$

(IV) $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

Pairwise Independence :-

Let A, B and C are any three events defined on Ω said to be pairwise independent if it satisfies the following conditions

(i) $P(A \cap B) = P(A) \cdot P(B)$

(ii) $P(B \cap C) = P(B) \cdot P(C)$

(iii) $P(A \cap C) = P(A) \cdot P(C)$

Note :- Mutual Independence implies pairwise independence but pairwise independence does not imply mutual independence

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Ex 1] Consider experiment of tossing an unbiased coin three times.

Let the A_i denote the event that a head

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turns up on i^{th} toss ; $i = 1, 2, 3$. Are event A_1, A_2, A_3 mutually independent ?

$\Rightarrow$ $\Omega = \{HHH, HHT, HTH, HTT, TTH, THH, THT, TTT\}$

A_1 = head turns up on 1^{st} toss

$A_1 = \{HHH, HHT, HTH, HTT\}$ $\quad P(A_1) = \frac{4}{8} = \frac{1}{2}$

A_2 = head turns up on 2^{nd} toss

$A_2 = \{HHH, HHT, THH, THT\}$ $\quad P(A_2) = \frac{4}{8} = \frac{1}{2}$

A_3 = head turns up on 3^{rd} toss

$A_3 = \{HHH, HTH, TTH, THH\}$ $\quad P(A_3) = \frac{4}{8} = \frac{1}{2}$

(i) $P(A_1 \cap A_2) = P(A_1) \cdot P(A_2)$

L.H.S $= P(A_1 \cap A_2)$

$(A_1 \cap A_2) = \{HHH, HHT\}$

$P(A_1 \cap A_2) = \frac{2}{8} = \frac{1}{4}$

R.H.S $= P(A_1) \cdot P(A_2)$

$= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

Here L.H.S = R.H.S

eq^n (i) is satisfied

$\therefore$ A_1 & A_2 are $\perp\!\!\!\perp$

(ii) $P(A_2 \cap A_3) = P(A_2) \cdot P(A_3)$

L.H.S $= P(A_2 \cap A_3)$

$A_2 \cap A_3 = \{HHH, THH\}$

$P(A_2 \cap A_3) = 2/8 = \frac{1}{4}$

$P(A_2) \cdot P(A_3) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$ = R.H.S

L.H.S = R.H.S

A_2 and A_3 are $\underline{\perp\!\!\!\perp}$

(iii) $P(A_1 \cap A_3) = P(A_1) \cdot P(A_3)$

L.H.S $= P(A_1 \cap A_3)$

$A_1 \cap A_3 = \{HHH, HTH\}$

$P(A_1 \cap A_3) = \frac{2}{8} = \frac{1}{4}$

$P(A_1) \cdot P(A_3) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$ = R.H.S

∴ L.H.S = R.H.S

(iv) $P(A_1 \cap A_2 \cap A_3) = P(A_1) \cdot P(A_2) \cdot P(A_3)$

L.H.S $= P(A_1 \cap A_2 \cap A_3)$

$A_1 \cap A_2 \cap A_3 = \{HHH\}$

$P(A_1 \cap A_2 \cap A_3) = \frac{1}{8}$

$P(A_1) \cdot P(A_2) \cdot P(A_3) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$ = R.H.S

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L.H.S = R.H.S

Since A_1, A_2 and A_3 satisfies all the 4 condit^{n}s we can say that they are mutually independent.

∴ Conditional Probability :-

Let A and B are any events defined on Ω the conditional probability of A given B is

$$\boxed{P\left(\frac{A}{B}\right) = \frac{P(A\cap B)}{P(B)}} \quad ; \quad P(B) > 0$$ important to mention

and Multiplication Theorem is,

$$\boxed{P(A\cap B) = P\left(\frac{A}{B}\right) \cdot P(B)}$$

Similarly, conditional probability of B given A is

$$P\left(\frac{B}{A}\right) = \frac{P(B\cap A)}{P(A)} \quad ; \quad P(A) > 0$$

and its Multiplication Theorem,

$$P(B\cap A) = P\left(\frac{B}{A}\right) \cdot P(A)$$

Ex 1:- A pair of fair dice is rolled. If the sum of 8 has appeared, find the probability that one of the dice shows 3.

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Ans ⇒ $\Omega = \{(1,1), (1,2), \ldots\ldots, (6,6)\}$

$n(\Omega) = 36$

A: Let A be the event where dice form sum of 8.

$A = \{(2,6), (3,5), (4,4), (5,3), (6,2)\}$

$n(A) = 5$

B: let A be event where dice has 3.

$B = \{(3,5), (5,3)\}$

$n(B) = 2$

$$P(B) = \frac{n(B)}{n(\Omega)} = \frac{2}{36} = \frac{1}{18}$$

$$\boxed{P(A \cap B) = \frac{2}{36} = \frac{1}{18}}$$

$$\boxed{P(A) = \frac{n(A)}{n(\Omega)} = \frac{5}{36}}$$

∴ Probability that one of the dice shows 3 $= P(B/A)$

$$= \frac{P(A \cap B)}{P(A)} = \frac{2/36}{5/36}$$

$$= \boxed{\frac{2}{5}}$$

Ex 8] A random experiment result in an integer outcomes between 1 to 10 (both inclusive) All number are equally likely. Let A be the event that an odd number occurrs and B be the event than a no. divisible by 3 occurrs obtain :-

(i) $P(A/B)$ (ii) $P(B/A)$

(iii) $P(A'/B)$ (iv) $P(A/B')$ (v) $P(A'/B')$

Ans ⇒

$\Omega = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$n(\Omega) = 10$

A : Odd number occurres

$A = \{1, 3, 5, 7, 9\}$

$n(A) = 5$

B : Number divisible by 3

$B = \{3, 6, 9\}$ $B' = \{1, 2, 4, 5, 7, 8, 10\}$

$n(B) = 3$ $n(B') = 7$

$P(A) = \frac{n(A)}{n(\Omega)} = \boxed{\frac{5}{10}}$

$P(B) = \frac{n(B)}{n(\Omega)} = \boxed{\frac{3}{10}}$ $P(B') = \boxed{\frac{7}{10}}$

$A' = \{2, 4, 6, 8, 10\}$

$n(A') = 5$ $P(A') = \boxed{5/10}$

$A \cap B = \{3, 9\}$

$n(A \cap B) = 2$

$P(A \cap B) = \dfrac{n(A \cap B)}{n(\Omega)} = \boxed{\dfrac{2}{10}}$

$A \cap B' = \{1, 5, 7\}$

$n(A \cap B') = 3$

$P(A \cap B') = \dfrac{n(A \cap B')}{n(\Omega)} = \boxed{\dfrac{3}{10}} =$

$A' \cap B = \{6\}$

$n(A' \cap B) = 1$

$P(A' \cap B) = \dfrac{n(A' \cap B)}{n(\Omega)} = \boxed{\dfrac{1}{10}}$

$A' \cap B' = \{2, 4, 8, 10\}$

$n(A' \cap B') = 4$

$P(A' \cap B') = \dfrac{n(A' \cap B')}{n(\Omega)} = \boxed{\dfrac{4}{10}}$

(i) $P\left(\dfrac{A}{B}\right) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{2/10}{3/10} = \dfrac{2}{3}$

(ii) $P\left(\dfrac{B}{A}\right) = \dfrac{P(A \cap B)}{P(A)} = \dfrac{2/10}{5/10} = \dfrac{2}{5}$

(iii) $P\left(\dfrac{A'}{B}\right) = \dfrac{P(A' \cap B)}{P(B)} = \dfrac{1/10}{3/10} = \dfrac{1}{3}$

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(iv) $P\left(\frac{A}{B'}\right) = \frac{P(A \cap B')}{P(B')} = \frac{3/10}{7/10} = \boxed{\frac{3}{7}}$

(v) $P\left(\frac{A'}{B'}\right) = \frac{P(A' \cap B')}{P(B')} = \frac{4/10}{7/10} = \boxed{\frac{4}{7}}$

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